

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

28th June 2019

Subject CP2A – Actuarial Modelling (Paper A)

Time allowed: 3 Hour 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

AT THE END OF THE EXAMINATION

Please return this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you. You are requested to save and submit the work before leaving the examination premises.

Examination instructions

1. *Mark allocations are shown in brackets.*
2. *Do save your work in Excel files on a regular basis.*
3. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*
4. *Please note that you should use your Roll Number and NOT your name on all of the material you submit for marking.*
5. *You would be provided with a digital platform for the examination which will show the time remaining for the examination at all times, once the exam starts. Once the exam duration is over, any work done on MS Excel will automatically get submitted and the platform will get frozen.*
6. *There will however be a 'Submit' button available. Ensure that you click the "Submit" button only when you have answered all questions including sub-parts and then final submission is made. If done erroneously at any point earlier, this step can't be un-done.*
7. *Students are required to submit all their work i.e. model, audit and summary using MS excel only.*

Exam Requirements

- 1) Read the background document, which describes the scenarios that need to be modelled and documented for this project. Technical assistance for the modelling work, should you require it, can be found in the additional guidance contained in this booklet.

No marks will be deducted for the use of this guidance.

- 2) Construct a spreadsheet model that produces the following calculations and charts. You should ensure that your spreadsheet contains appropriate self-checks and that you have performed robust reasonableness checks at each stage of your calculations.
 - i) Calculate the projected number of Puranium, Uranium and total rods over a three year period on a monthly basis. (4)
 - ii) Produce a suitable chart for the projection in part (i) showing the projection for each material's rods as well as the total rods projection. (2)
 - iii) Using the information in the background and base scenario:
 - a) Construct a monthly projection of the locker utilisation over a three year period for each radioactive material. (4)
 - b) Determine the monthly projection of surplus/shortage of lockers over a three-year period at an aggregate level. Also produce a chart showing the projection of the surplus lockers for the bank as a whole. (8)
 - c) Determine the outgo for Edvert depending upon the locker projections showing a surplus or a shortage. (6)
 - iv) Repeat part (i) and (iii) for the request given in Scenario 2. (2)
 - v) Repeat part (i) and (iii) for the request given in Scenario 3. Determine the number of Uranium purchases every month that will result in nil surplus at the end of three years. (3)
 - vi) Produce a suitable chart showing the total outgo for Edvert across the three scenarios, split by type of outgo i.e. renovation cost, shortage penalty and maintenance cost of surplus lockers. (4)

Marks available for spreadsheet model and checks:

- Accurate completion of above modelling steps and data checks. (33)
- Demonstration of good modelling technique and practice. (7)
- Other (non-data) checks. (7)

[47]

Produce an audit trail for your spreadsheet model which includes the following aspects:

- purpose of the model
- data and assumptions used
- methodology, i.e. description of how each calculation stage in the model has been produced
- explanation of the checks performed

You should ensure that your audit trail is suitable for both a senior actuary, who has been asked to approve your work, and a fellow student, who has been asked to peer review and correct your model, or to continue work on it, or to use it again for a similar purpose in the future.

Marks available for audit trail:

Audit approach

Fellow student can review and check methods used in the model	(8)
Senior actuary can scrutinise and understand what has been done	(8)
Written in clear English	(4)
Written in a logical order	(3)

Audit content

All steps clearly explained	(7)
Clear signposting included throughout	(5)
Statement of assumptions made	(5)
All model steps accurately covered	(13)
	[53]

Background

On a sparsely populated alien planet called Marsh, Edvert's ancestors have historically run a bank that provides private specialised lockers to keep rare radioactive elements safe and far from Planet Pearth, where the population resides. The bank has a fixed capacity in terms of number of lockers that can be made available. Edvert has now inherited the management of this bank from his father.

The bank specialises in two kind of radioactive materials (kept in the form of rods) –Puradium and Uradium.

Following is the key information given:

- Puradium has a tendency to multiply itself at a certain rate while in possession; Uradium does not multiply itself, however both the radioactive elements certainly decay at a certain rate despite utmost care provided in the specialised lockers.
- His father had signed a two-year agreement where every month the bank will receive 11 (Eleven) Uradium rods from planet Pearth. The agreement was executed six months ago.
- One locker accommodates only one kind of radioactive rod otherwise they may react together and explode.
- The lockers suffer from the radiation of the radioactive rods and hence there is a cost of renovation of lockers.
- The planet Pearth regulation levies a heavy penalty on bank owners if they do not have enough lockers in their bank to keep all the radioactive rods.

Edvert intends to keep the bank for the next three years and is worried whether there will be enough lockers to keep all of the rods in the locker over this period.

He has done his survey and has noted the following:

- Puradium rods in the Bank at present– 2,000 rods
- Uradium rods in the Bank at present – 200 rods
- Total locker capacity of the Bank –2,500
- Cost to renovate an existing locker suffered from radiations– M\$50,000 per locker. (M\$ is Marshian Dollar). This cost is incurred only when the locker utilisation changes from one element to the other.
- The penalty levied by the Pearth Government is M\$105,000 per locker per month for the number of lockers that are short.
- The multiplication and decay rates of the radioactive materials (provided in a separate excel sheet; these are annual rates).

While the shortage of lockers attracts a penalty, the surplus lockers also incur a maintenance cost of M\$15,000 per month. Edvert is not worried about the maintenance cost of occupied lockers or other expenses like staff salaries etc. as the revenue from other bank activities is expected to cover these.

Edvert is interested to know how the next three years are likely to progress in terms of locker utilization and his outgo liability due to maintenance, renovation or penalty.

Base Scenario

You are a Consultant in a Company called BIG4. Your boss has asked you to project the number of rods for the next three years and find the total number of rods, for each radioactive material, at the end of three years. He has also asked you to calculate the surplus/shortage in the number of lockers every month and the outgo Edvert is expected to incur from renovating the lockers, maintaining the surplus lockers and funding the Government penalty from any shortage in the number of lockers.

Scenario 2

The Government of Planet Pearth plans to come with a new regulation which may prohibit further production of Uranium rods as they are very risky to the environment in comparison to the Puranium.

Edvert is concerned of the possibility of implementation of the regulation and asks your boss to calculate his outgo if there are no further Uranium rods received by the bank.

Scenario 3

Edvert wants to optimise the number of lockers and asks your boss to calculate the monthly number of Uranium rods to purchase which will ensure no shortage and the minimum surplus lockers at the end of 36 months.

Multiplication and decay rates have been given in a separate excel sheet.

Note: When projecting the rods do not show the number of rods with decimal places. You are recommended to round your answers to the nearest integer at every appropriate intermediate stage.

Additional Guidance

Assistance with the following aspects of the modelling is given in the Additional Guidance sheets included within this examination insert. [No penalty marks are deducted for use of this guidance]

Additional Guidance A: Projecting the rod and surplus/shortage of lockers.

Additional Guidance B: Determining the number of Uranium rods required under Scenario 3.

Additional Guidance A: Projecting the rods and surplus/shortage of lockers.

These calculations should be performed separately for the two radioactive material.

Count of rods at Start

For $t=1$, this is set equal to the number of rods as per the start data received by Edvert.

For $t=2$ onwards, $\text{Rods_Start}(t) = \text{Rods_End}(t-1)$

Rods added during the month

$\text{New_Rods_Added}(t) = \text{Rods_Start}(t) * \text{monthly reproduction rate applicable, in case of Puranium.}$

$\text{New_Rods_Added}(t) = \text{per month purchase agreed if the period of purchase contract is still valid, in case of Uranium.}$

The result should be rounded to nearest integer using ROUND Excel function.

Rods exiting during the month

$\text{Rods_Exiting}(t) = \{\text{Rods Start}(t) + \text{New_Rods_Added}(t)\} * \text{monthly decay rate.}$

The result should be rounded to nearest integer using ROUND Excel function.

Rods at end of month

$\text{Rods_End}(t) = \text{Rods Start}(t) + \text{New_Rods_Added}(t) - \text{Rods_Exiting}(t)$

Surplus and Shortage of Lockers

$\text{Surplus_Lockers}(t) = \text{MAX}(\text{Rods_Start}(t) - \text{Rods_End}(t), 0)$

$\text{Shortage_Lockers}(t) = \text{MAX}(\text{Rods_End}(t) - \text{Rods_Start}(t), 0)$

Using the above Rods projection counts and Lockers projection counts, aggregate calculations at the Bank level for surplus/shortage of Lockers should then be performed.

Additional Guidance B: Determining the number of Uranium required under Scenario 3.

Goal Seek should be used to determine the number of Uranium purchases.

For this, the aggregate count of surplus lockers at the end of 36 months should be calculated and goal seeked to be zero by changing the number of Uranium purchases for the remaining contract term.

Your answer may then be reduced to the nearest integer as rods count must not be in fractions and Edvert does not want any shortage.

Link for Data Set:

https://actuariesindia.org/sites/default/files/2022-09/Rate_Data_CP2A_updated.xls
